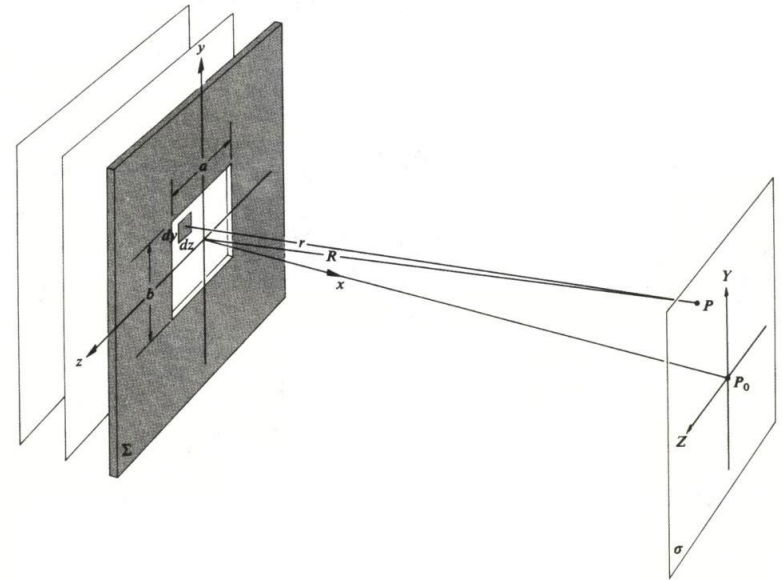


The rectangular aperture

$$\tilde{E} = \frac{\epsilon_A e^{i(\omega t - kR)}}{R} \int_{-\frac{b}{2}}^{\frac{b}{2}} e^{ikYy} dy \int_{-\frac{a}{2}}^{\frac{a}{2}} e^{ikZz} dz$$

$$\tilde{E} = \frac{A\epsilon_A e^{i(\omega t - kR)}}{R} \left(\frac{\sin \alpha'}{\alpha'} \right) \left(\frac{\sin \beta'}{\beta'} \right)$$

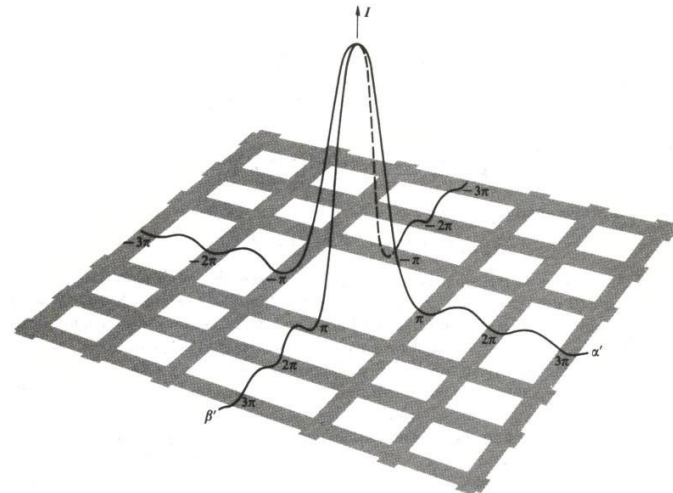
$$\alpha' \equiv kaZ/2R \quad \beta' \equiv kbY/2R$$



Irradiance:

$$I(Y, Z) = I(0) \left(\frac{\sin \alpha'}{\alpha'} \right)^2 \left(\frac{\sin \beta'}{\beta'} \right)^2$$

$I(0)$: the irradiance at $Y=0$ and $Z=0$.



The circular aperture

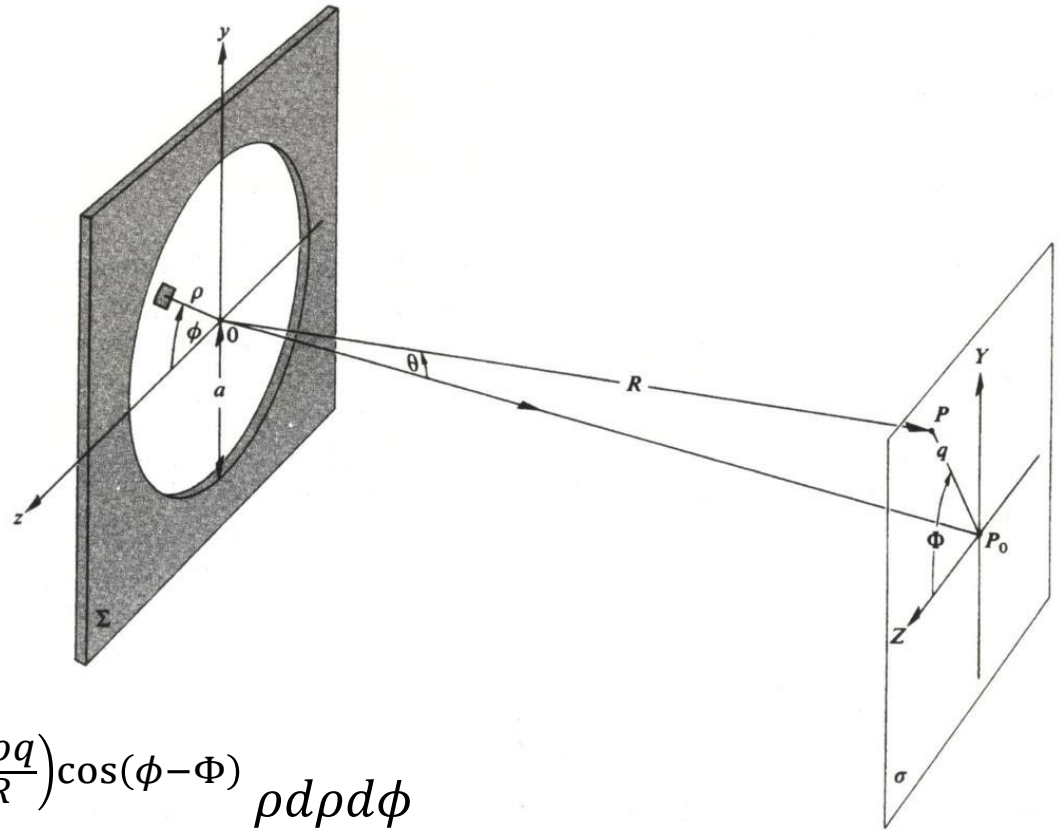
Optical disturbance at P:

$$\tilde{E} = \frac{\epsilon_A e^{i(\omega t - kR)}}{R} \int_{\text{Aperture}} e^{ik(Yy + Zz)} dS$$

For a circular opening:

$$z = \rho \cos \phi \quad y = \rho \sin \phi$$

$$Z = q \cos \Phi \quad Y = q \sin \Phi$$



$$\tilde{E} = \frac{\epsilon_A e^{i(\omega t - kR)}}{R} \int_{\rho=0}^a \int_{\phi=0}^{2\pi} e^{i\left(\frac{k\rho q}{R}\right)\cos(\phi-\Phi)} \rho d\rho d\phi$$

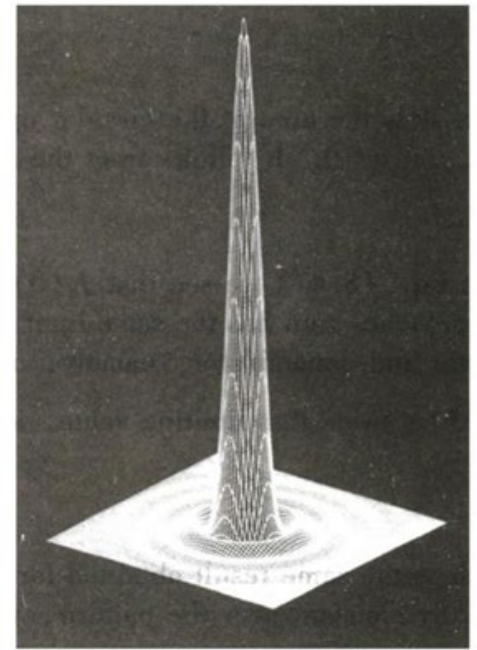
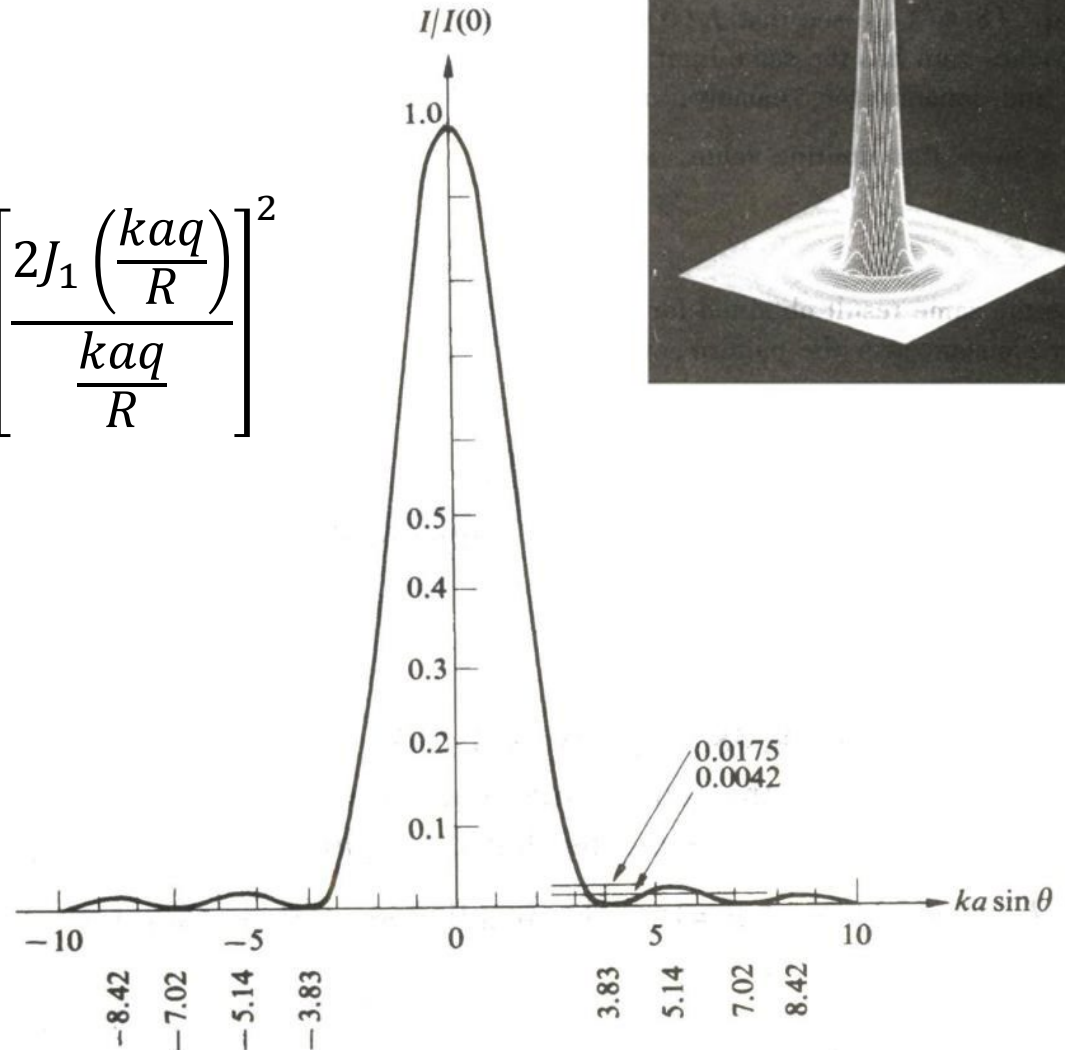
$$\tilde{E} = \frac{\epsilon_A e^{i(\omega t - kR)}}{R} 2\pi a^2 \left(\frac{R}{kaq} \right) J_1 \left(\frac{kaq}{R} \right)$$

The irradiance at P:

$$I = \frac{2\epsilon_A^2 A^2}{R^2} \left[\frac{J_1 \left(\frac{kaq}{R} \right)}{\frac{kaq}{R}} \right]^2 = I(0) \left[\frac{2J_1 \left(\frac{kaq}{R} \right)}{\frac{kaq}{R}} \right]^2$$

$$= I(0) \left[\frac{2J_1(ka \sin \theta)}{ka \sin \theta} \right]^2$$

$I(0)$: the irradiance at P_0 .



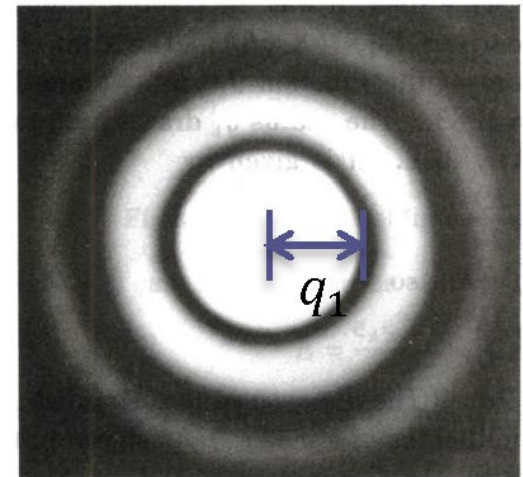
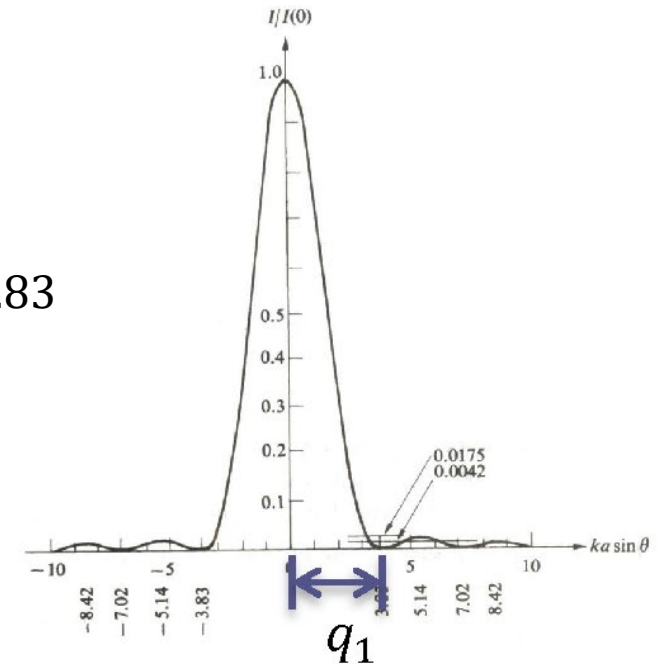
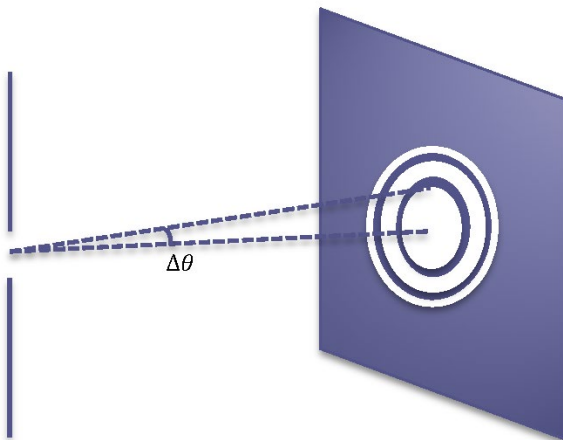
Airy disk: the towering central maximum corresponds to a high-irradiance circular spot.

The radius of Airy disk: $J_1(u) = 0$ when $u = \frac{kaq}{R} = 3.83$

$$q_1 = \frac{3.83 R}{ka} = 1.22 \frac{R\lambda}{2a} \approx 1.22 \frac{f\lambda}{D}$$

The angular measure:

$$\Delta\theta \approx \sin \Delta\theta = \frac{q_1}{f} = 1.22\lambda/D$$



Airy rings using a 0.5-mm hole diameter

Resolution of Imaging Systems

Example: two stars seen through the objective lens of a telescope (the entrance pupil corresponds to the diffracting aperture)

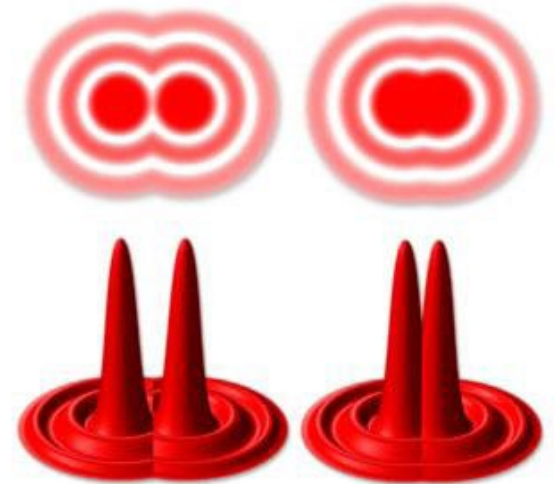
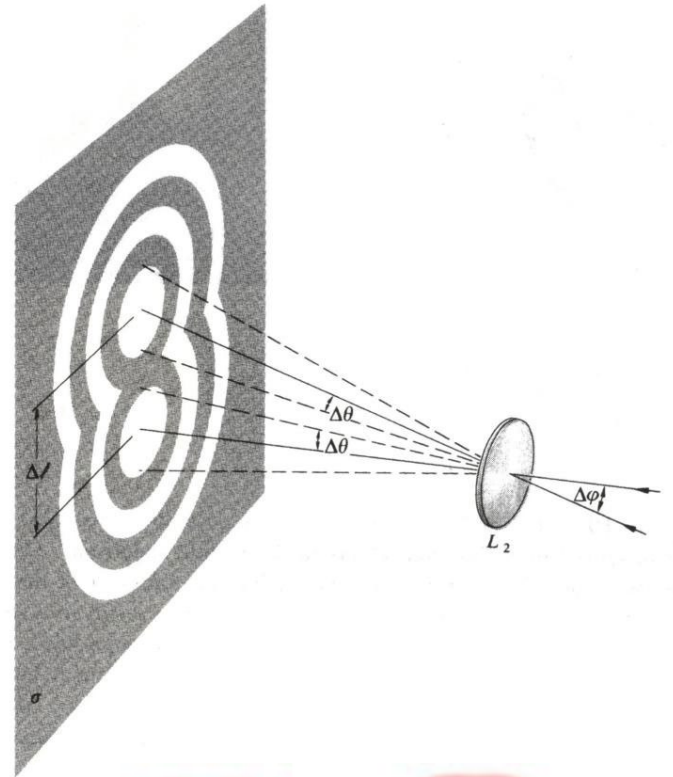
If Lord Rayleigh's criterion is applied, the stars are said to be *just resolved* when the center of one Airy disk falls on the first minimum of the Airy pattern of the other star.

The minimum resolvable angular separation (Lord Rayleigh's criterion):

$$(\Delta\varphi)_{min} = \Delta\theta = 1.22\lambda/D$$

Improve the revolution:

- ☐ Decrease the wavelength;
- ☐ Increase the diameter of the objective lens.



Lord Rayleigh's criterion

Grating Spectroscopy

The angular distance between the zeros on either side of a principal maximum.

$$\Delta\alpha = 2\pi/N$$

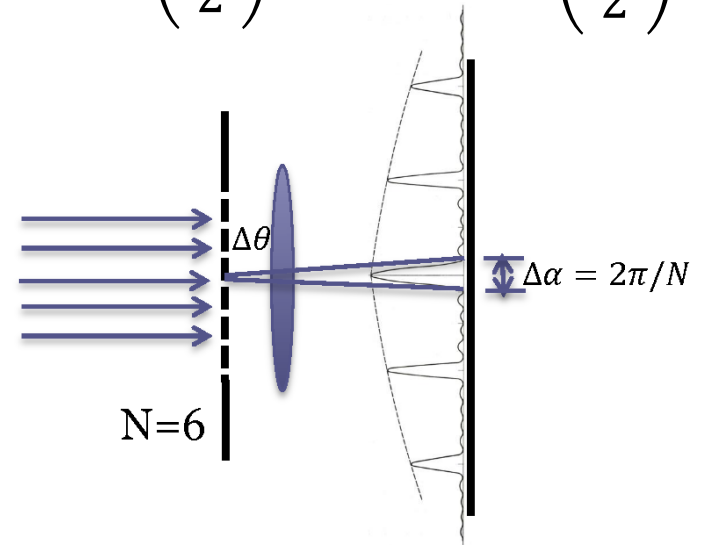
At oblique incidence:

$$\alpha = (ka/2)(\sin\theta - \sin\theta_i)$$

$$\Delta\alpha = \left(\frac{ka}{2}\right) \cos\theta (\Delta\theta) = 2\pi/N$$

$$I(\theta) = I_0 \left(\frac{\sin\beta}{\beta}\right)^2 \left(\frac{\sin N\alpha}{\sin\alpha}\right)^2$$

$$\beta \equiv \left(\frac{kb}{2}\right) \sin\theta \quad \alpha \equiv \left(\frac{ka}{2}\right) \sin\theta$$



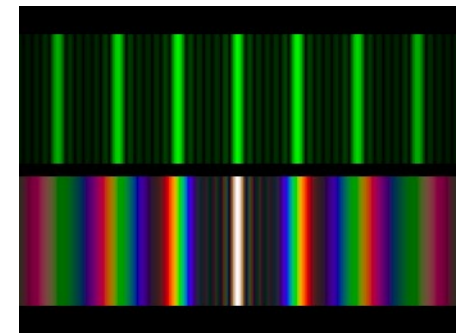
Angular width of a line: $\Delta\theta = 2\lambda/(Na \cos\theta_m)$

↙ Grating width

Angular dispersion: the difference in angular position corresponding to a difference in wavelength:

$$\frac{d\theta}{d\lambda} = \frac{m}{a \cos\theta_m}$$

$$\begin{aligned} a(\sin\theta_m - \sin\theta_i) &= m\lambda \\ a \cos\theta d\theta &= m d\lambda \end{aligned}$$

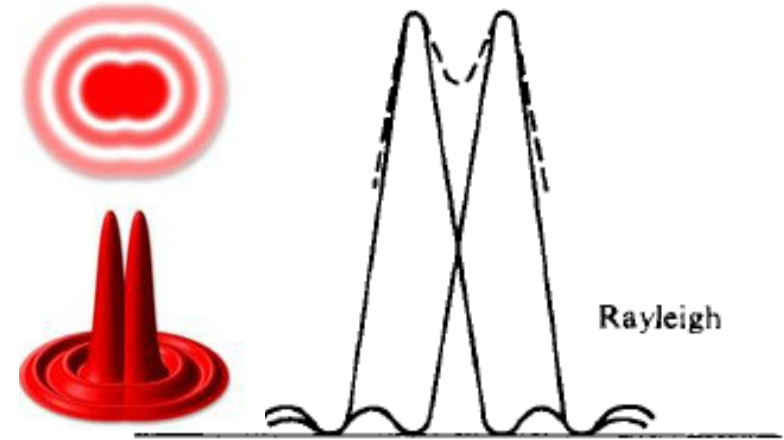


Lord Rayleigh's criterion: the principal maximum of one coincide with the first minimum of the other, at the limit of resolution, the angular separation is half the linewidth, $d\theta = \frac{\Delta\theta}{2}$

$$(\Delta\theta)_{min} = \lambda / (Na \cos \theta_m)$$

$$\frac{d\theta}{d\lambda} = \frac{m}{a \cos \theta_m}$$


$$(\Delta\theta)_{min} = (\Delta\lambda)_{min} m / (a \cos \theta_m)$$



Lord Rayleigh's criterion

The chromatic resolving power (色分辨本领) of a spectrometer:

$$\mathcal{R} \equiv \frac{\lambda}{(\Delta\lambda)_{min}} = mN = \frac{Na(\sin \theta_m - \sin \theta_i)}{\lambda}$$

Example: A grating 6 inches wide and containing 15,000 lines per inch will have a total of 9×10^4 lines and a resolving power, in the second order, of $R = 1.8 \times 10^5$. In the vicinity of 540 nm, the grating could resolve a wavelength difference of 0.003 nm.

9.3 Fresnel Diffraction

Huygens-Fresnel Principle: at any instant, every point on the primary wavefront is envisioned as a continuous emitter of spherical secondary wavelet. It means each wavelet radiate uniformly in all direction.

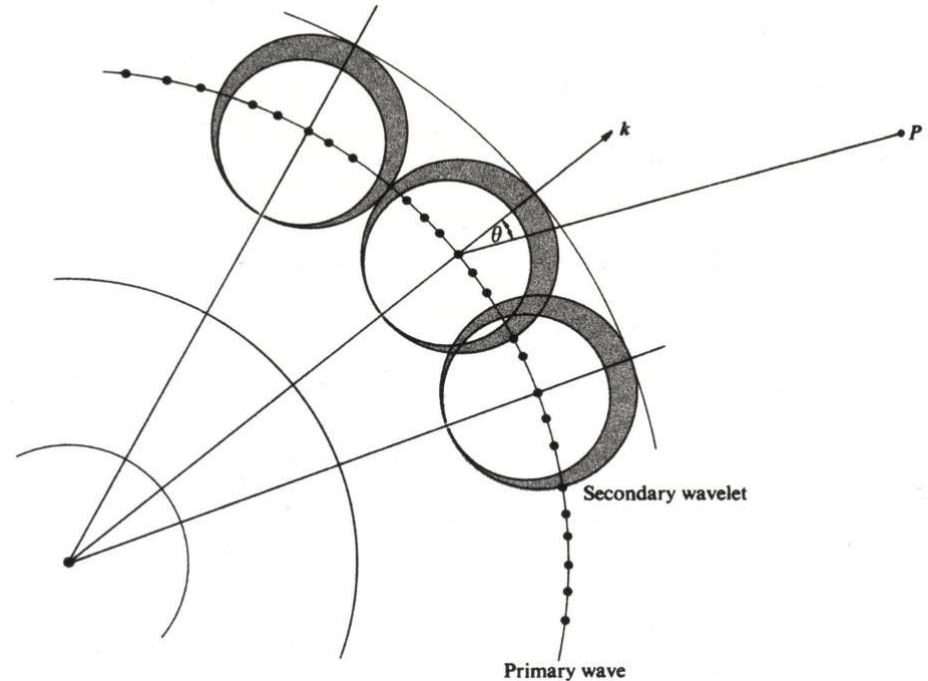
In experiment, there is no reverse wave traveling back toward the source.

Obliquity (inclination factor): it describes the directionality of the secondary emissions.

Kirchhoff formulation:

$$K(\theta) = \frac{1}{2}(1 + \cos \theta)$$

θ : the angle made with the normal to the primary wavefront (k).



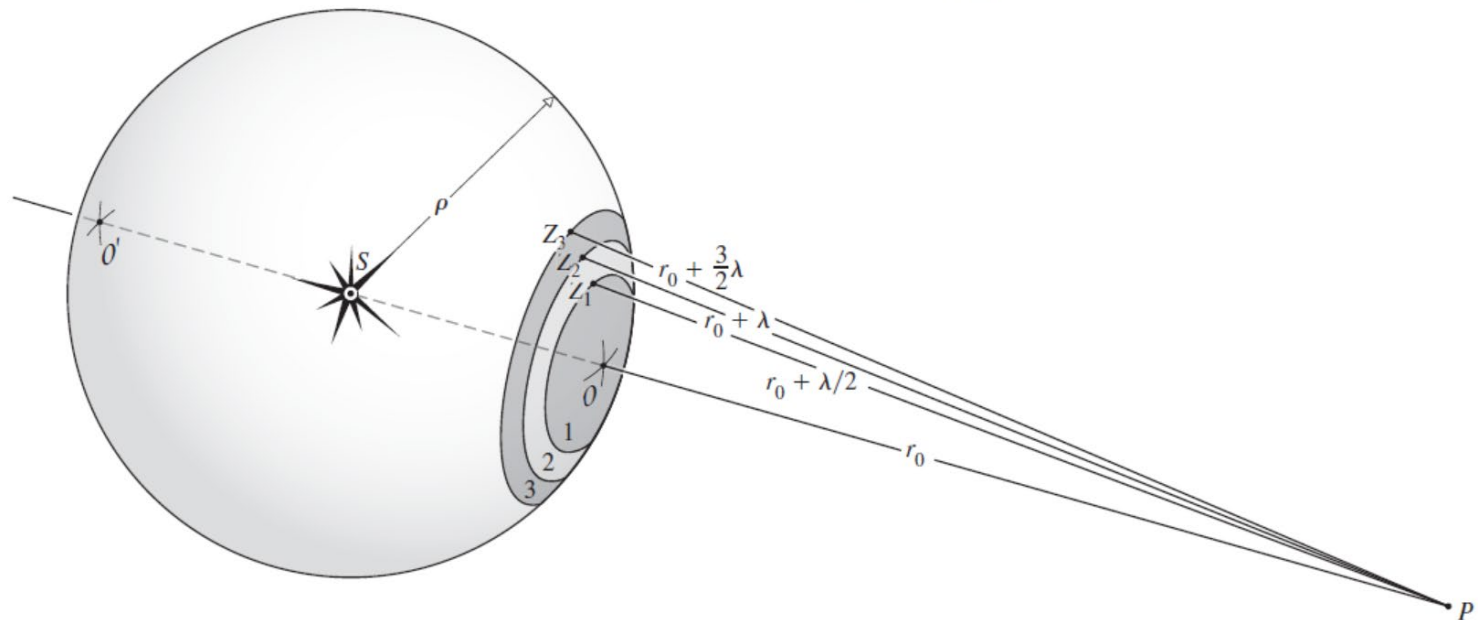
Case: the free propagation of a spherical monochromatic wave emitted from a point source, S.

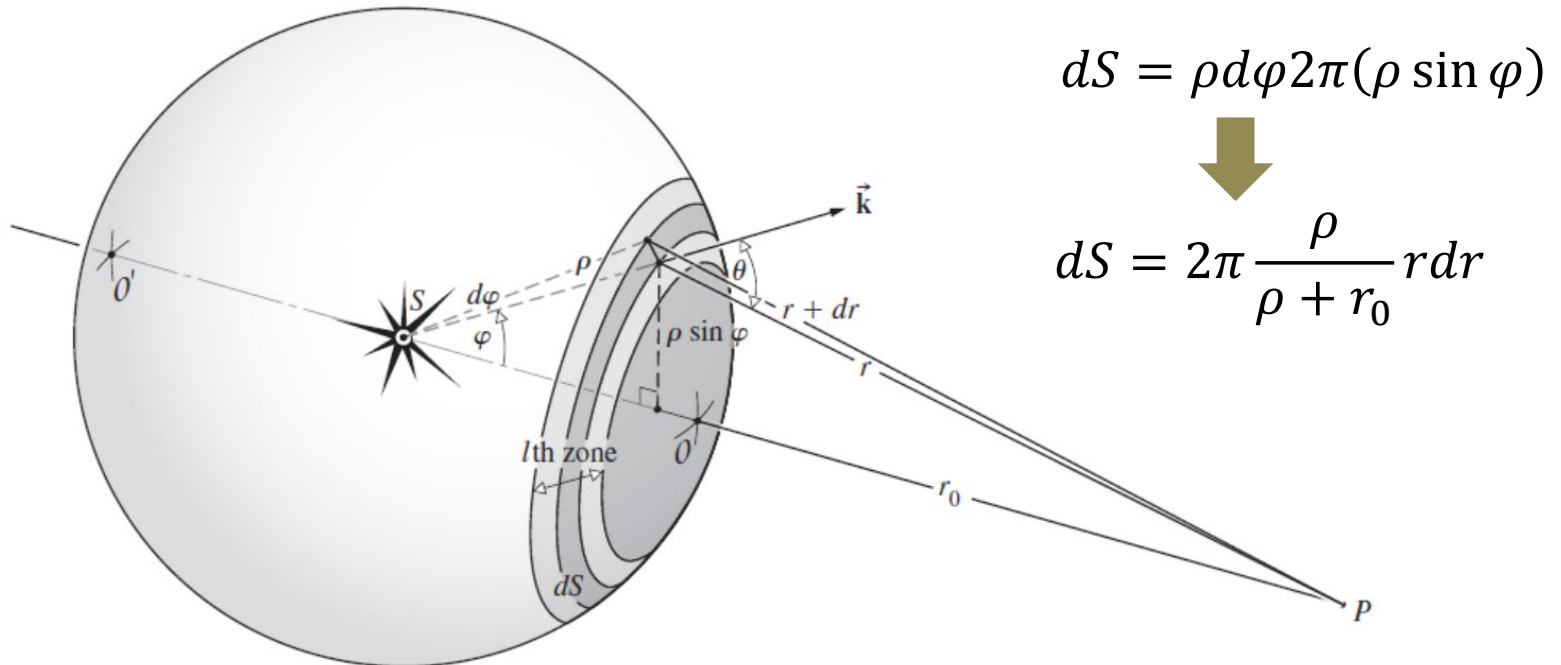
The spherical surface: primary wavefront at some arbitrary time t' emitted from S at $t=0$.

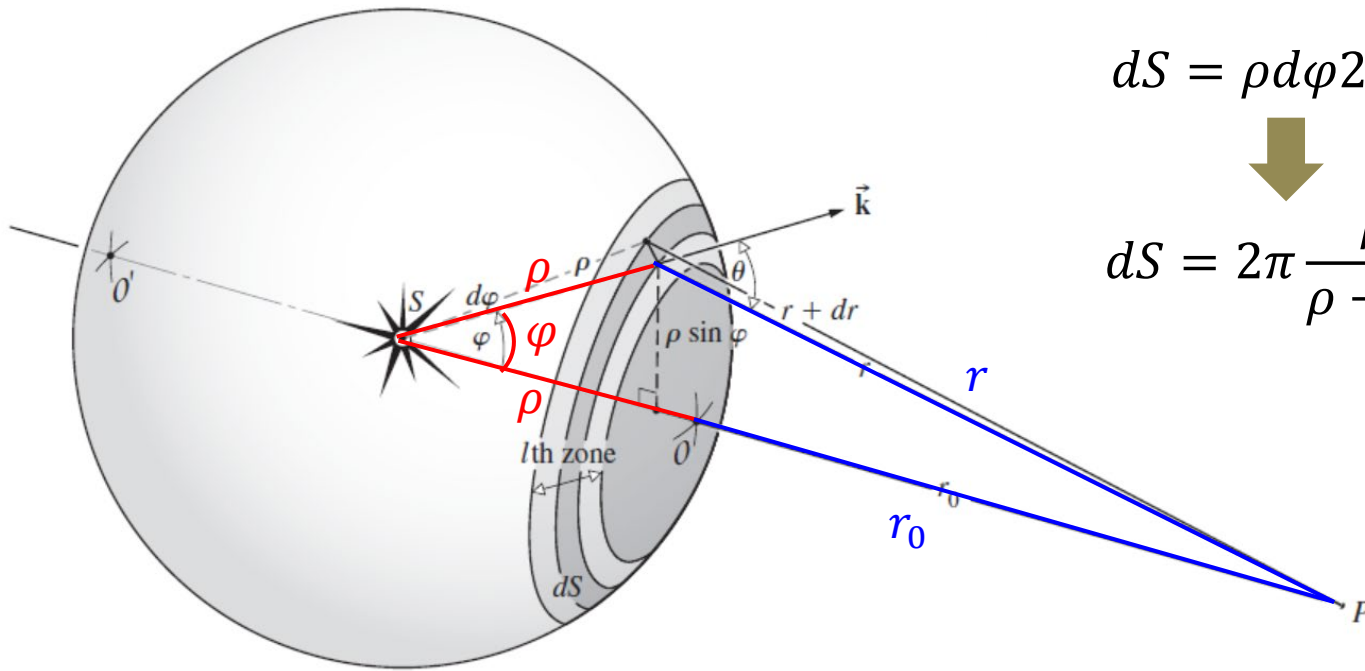
The **disturbance**:

$$E = \frac{\epsilon_0}{\rho} \cos(\omega t' - k\rho)$$

Fresnel (half-period zones): divide the wavefront into a number of annular region with the radius of $r_0 + \lambda/2, r_0 + \lambda, r_0 + 3\lambda/2, \dots$, centered at P.







$$dS = \rho d\varphi 2\pi(\rho \sin \varphi)$$



$$dS = 2\pi \frac{\rho}{\rho + r_0} r dr$$

$$r^2 = \rho^2 + (\rho + r_0)^2 - 2\rho(\rho + r_0) \cos \varphi$$

Keep ρ and r_0 unchanged, $2r dr = 2\rho(\rho + r_0) \sin \varphi d\varphi$

$$dS = \rho d\varphi 2\pi(\rho \sin \varphi) = 2\pi \frac{\rho}{\rho + r_0} r dr$$

The disturbance arriving at P from the l^{th} zone:

$$E_l = K_l 2\pi \frac{\epsilon_A \rho}{\rho + r_0} \int_{r_{l-1}}^{r_l} \cos[\omega t - k(\rho + r)] dr$$

$$r_l = r_0 + l\lambda/2$$

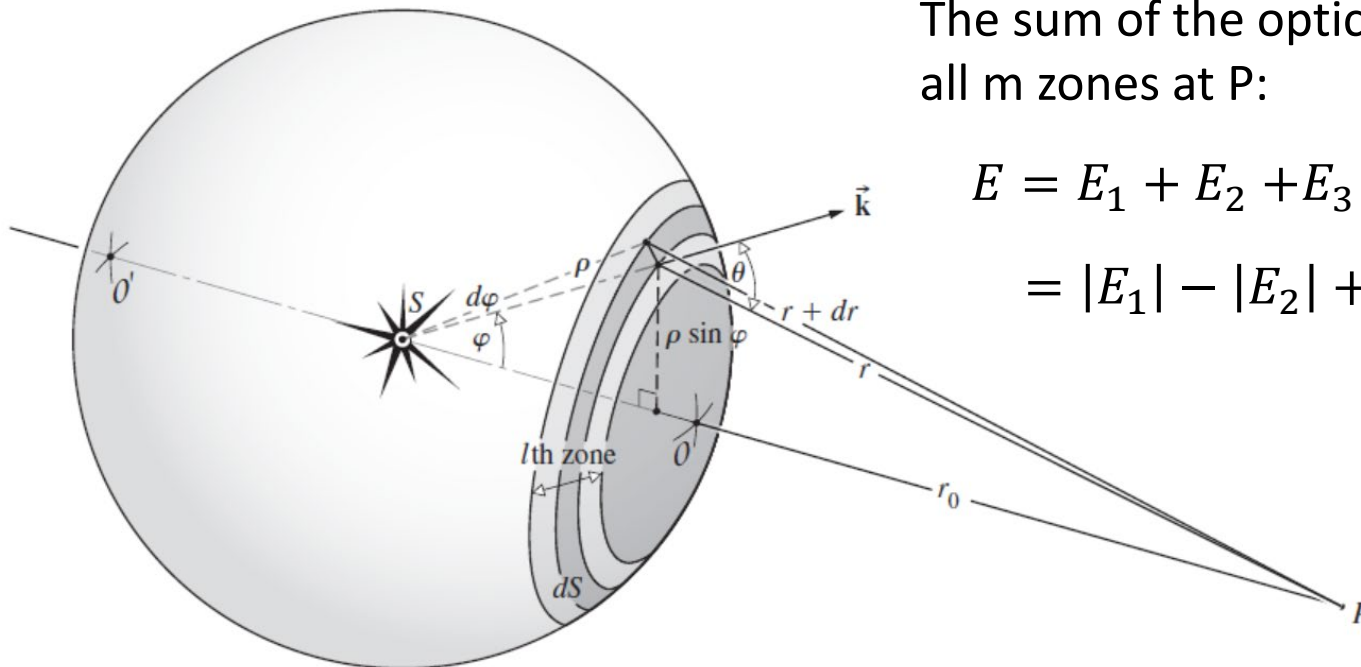
$$r_{l-1} = r_0 + (l-1)\lambda/2$$

$$E_l = (-1)^{l+1} \frac{2K_l \epsilon_A \rho \lambda}{(\rho + r_0)} \sin[\omega t - k(\rho + r_0)]$$

The sum of the optical disturbances from all m zones at P:

$$E = E_1 + E_2 + E_3 + \cdots + E_m$$

$$= |E_1| - |E_2| + |E_3| - \cdots + |E_m|$$



Case I: if m is odd, the sum of the optical disturbances from all m zones at P:

$$E = |E_1| - |E_2| + |E_3| - \cdots + |E_m|$$

$$E = \frac{|E_1|}{2} + \left(\frac{|E_1|}{2} - |E_2| + \frac{|E_3|}{2} \right) + \left(\frac{|E_3|}{2} - |E_4| + \frac{|E_5|}{2} \right) + \cdots + \left(\frac{|E_{m-2}|}{2} - |E_{m-1}| + \frac{|E_m|}{2} \right) + \frac{|E_m|}{2} \quad (\text{I})$$

$$E = |E_1| - \frac{|E_2|}{2} - \left(\frac{|E_2|}{2} - |E_3| + \frac{|E_4|}{2} \right) - \left(\frac{|E_4|}{2} - |E_5| + \frac{|E_6|}{2} \right) + \cdots + \left(\frac{|E_{m-3}|}{2} - |E_{m-2}| + \frac{|E_{m-1}|}{2} \right) - \frac{|E_{m-1}|}{2} + |E_m| \quad (\text{II})$$

$$\text{If } |E_l| > (|E_{l-1}| + |E_{l+1}|)/2$$

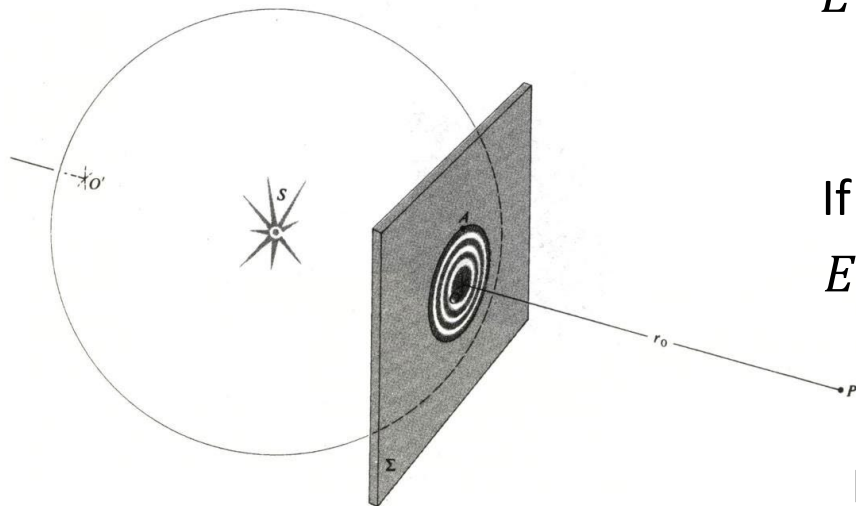
$$\begin{aligned} \text{Formula I: } E &< \frac{|E_1|}{2} + \frac{|E_m|}{2} \\ \text{Formula II: } E &> |E_1| - \frac{|E_2|}{2} - \frac{|E_{m-1}|}{2} + |E_m| \xrightarrow{|E_1| \approx |E_2|, |E_{m-1}| \approx |E_m|} \frac{|E_1|}{2} + \frac{|E_m|}{2} \end{aligned} \quad \left. \vphantom{\begin{aligned} \text{Formula I: } E &< \frac{|E_1|}{2} + \frac{|E_m|}{2} \\ \text{Formula II: } E &> |E_1| - \frac{|E_2|}{2} - \frac{|E_{m-1}|}{2} + |E_m| \end{aligned}} \right\} E \approx \frac{|E_1|}{2} + \frac{|E_m|}{2}$$

$$\text{If } |E_l| < (|E_{l-1}| + |E_{l+1}|)/2$$

$$\begin{aligned} \text{Formula I: } E &> \frac{|E_1|}{2} + \frac{|E_m|}{2} \\ \text{Formula II: } E &< |E_1| - \frac{|E_2|}{2} - \frac{|E_{m-1}|}{2} + |E_m| \xrightarrow{|E_1| \approx |E_2|, |E_{m-1}| \approx |E_m|} \frac{|E_1|}{2} + \frac{|E_m|}{2} \end{aligned} \quad \left. \vphantom{\begin{aligned} \text{Formula I: } E &> \frac{|E_1|}{2} + \frac{|E_m|}{2} \\ \text{Formula II: } E &< |E_1| - \frac{|E_2|}{2} - \frac{|E_{m-1}|}{2} + |E_m| \end{aligned}} \right\} E \approx \frac{|E_1|}{2} + \frac{|E_m|}{2}$$

Circular Apertures

A monochromatic spherical wave impinging on a screen containing a small hole.



The total disturbance at P, if m is even:

$$E = (|E_1| - |E_2|) + (|E_3| - |E_4|) + \dots + (|E_{m-1}| - |E_m|) \approx 0$$

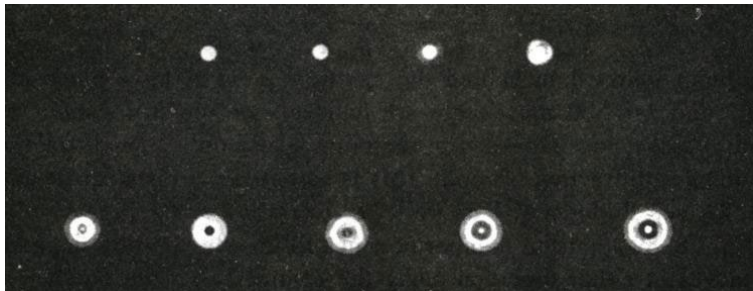
If m is odd

$$E = |E_1| - (|E_2| - |E_3|) - (|E_4| - |E_5|) - \dots - (|E_{m-1}| - |E_m|) \approx |E_1|$$

Note:

it is roughly twice the amplitude of the unobstructed wave

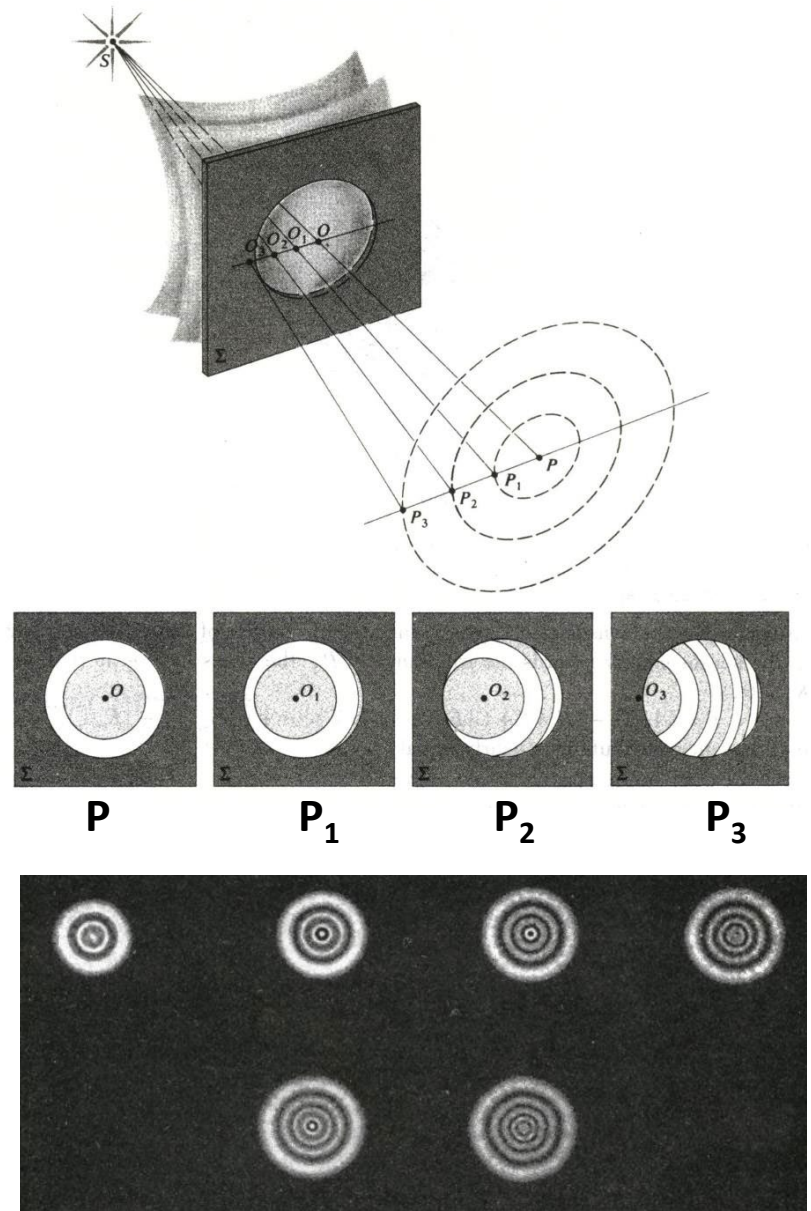
If m is not an integer, the irradiance at P is somewhere between zero and its maximum value



Diffraction patterns for circular apertures of increasing size

The rest of the pattern: assume that two complete zones fill the aperture

- At P: $E \approx 0$
- At P₁: the second zone has been partially obscured and the third begins to show. $E \neq 0$
- At P₂: a good fraction of the second zone is hidden, the third is even more evident. The contribution from the first and third zones are in-phase. A bright spot
- At P₃: the first zone is partially uncovered. The last few contributing segments is negligible. Further into the shadow, the entire first zone is obscured, the last terms are negligible. Darkness



Diffraction patterns for circular apertures of increasing size

The number of zones in a given aperture

The area of each zone:

$$dS = 2\pi \frac{\rho}{\rho + r_0} r dr \Rightarrow A = \frac{\rho}{(\rho + r_0)} \pi r_0 \lambda$$

The number zones within the aperture with a radius R.

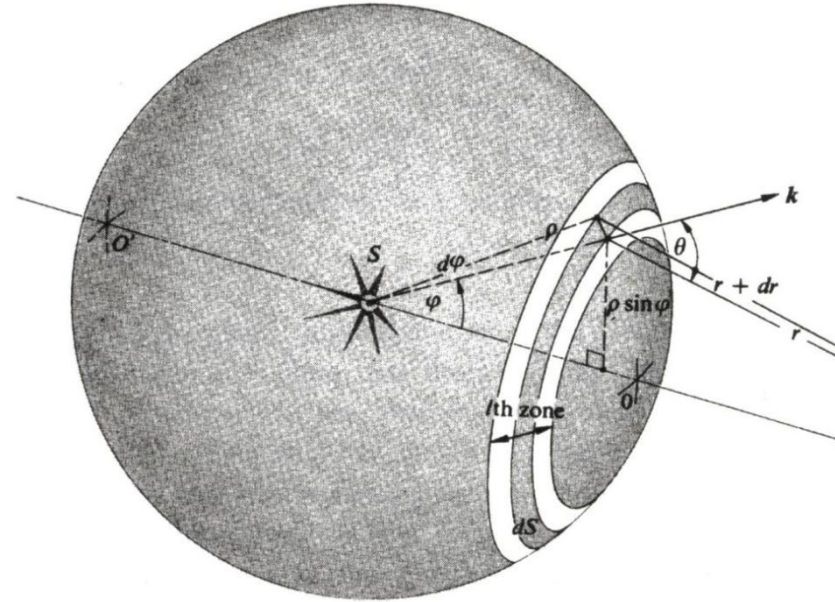
$$N = \frac{\pi R^2}{A} = \frac{(\rho + r_0) R^2}{\rho r_0 \lambda}$$

Example: a point source 1m behind the aperture ($\rho \approx 1m$), a plane of observation 1m in front of aperture ($r_0 = 1m$).

$$\lambda = 500 \text{ nm}$$

$$R = 1 \text{ mm}; 4 \text{ zones}$$

$$R = 1 \text{ cm}; 400 \text{ zones}$$



$$N \propto \frac{1}{r_0}$$

As P moves in either direction along the central axis, the number of uncovered zones oscillates between odd and even integers. As a result, the irradiance goes through a series of maxima and minima.

Circular Obstacles

When a circular obstacle is inserted between a light source and a screen P, suppose the first l zones are blocked off. The electrical field at P:

$$E = |E_{l+1}| - |E_{l+2}| + \cdots \pm |E_m|$$
$$\approx \frac{|E_{l+1}|}{2} \pm \frac{|E_m|}{2}$$

$K_m \rightarrow 0$, E_m approaches zero.

$$E \approx \frac{|E_{l+1}|}{2}$$

The irradiance on the central axis is only slightly less than that of the unobstructed wave ($E \approx \frac{|E_1|}{2}$).

There is a bright spot everywhere along the central axis except immediately behind the circular obstacle (because θ increases, $K_m \rightarrow 0$).

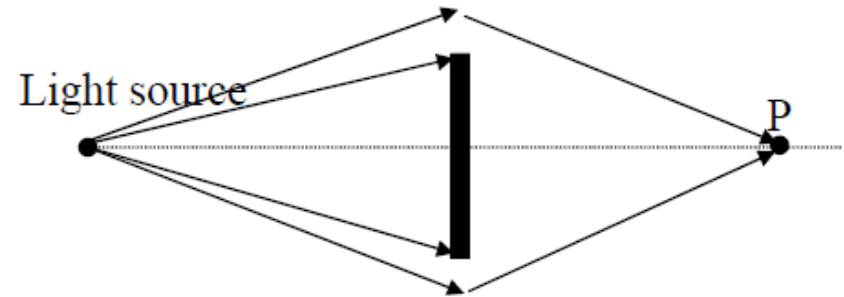
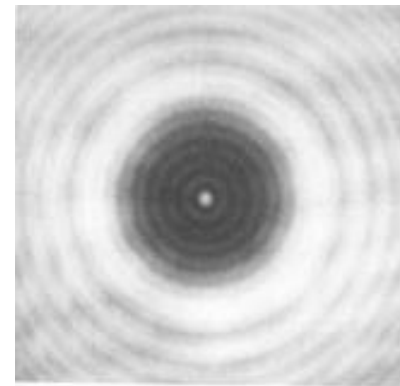
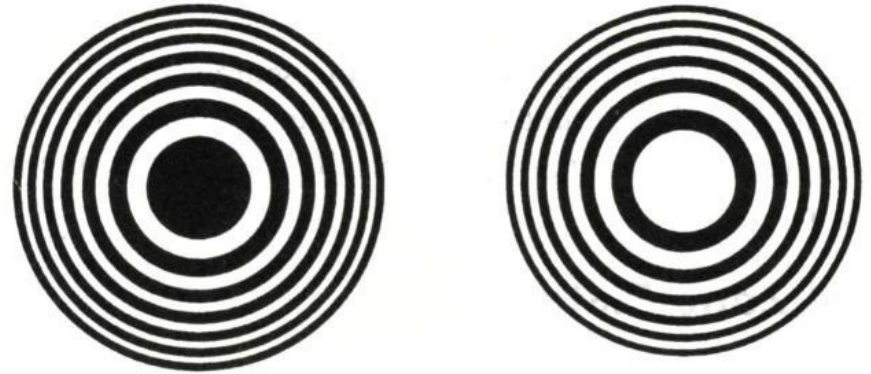


Fig 8.20 Circular Obstacle



The Fresnel Zone Plate

Zone plate: a screen that alter light, either in amplitude or phase, coming from every other half-period zone.



Example: a zone plate that passes only the first 20 odd zones and obstructs the even zone.

$$E = E_1 + E_3 + E_5 + \cdots + E_{39}$$

The disturbance at P: $E \approx 20E_1$

For an unobstructed wavefront: $E \approx \frac{1}{2}E_1$

The irradiance has been increased by a factor of 1600.

Calculate the radii of the zones

A wave traveling the path $S-Am-P$ must have the optical path difference by $m\lambda/2$ with a wave that travels the path $S-O-P$. (Am is the outer edge of the m th zone)

$$(\rho_m + r_m) - (\rho_0 + r_0) = m\lambda/2$$

Because

$$\rho_m = (R_m^2 + \rho_0^2)^{1/2} \quad r_m = (R_m^2 + r_0^2)^{1/2}$$

Expand in binomial series

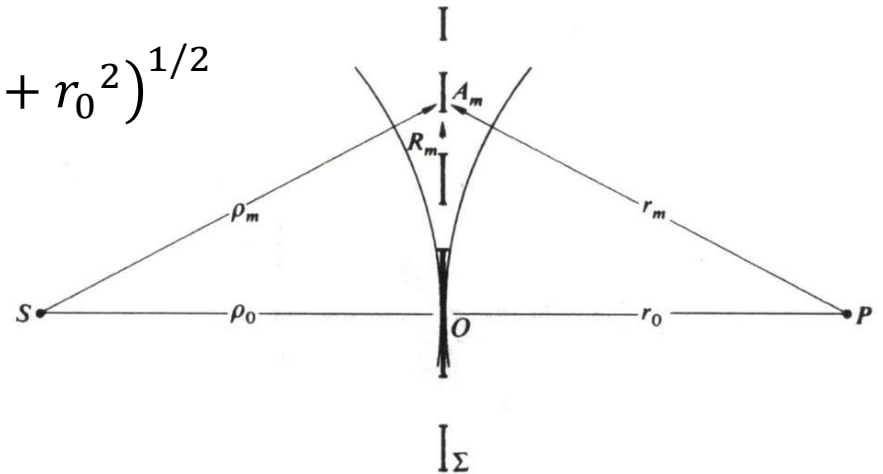
$$\rho_m = \rho_0 + \frac{R_m^2}{2\rho_0} \quad r_m = r_0 + \frac{R_m^2}{2r_0}$$

We have
$$\left(\frac{1}{\rho_0} + \frac{1}{r_0} \right) = \frac{m\lambda}{R_m^2}$$

When $\rho_0 \rightarrow \infty$, $R_m^2 = mr_0\lambda$ which has the same form of thin lens equation.

$$f_1 = \frac{R_m^2}{m\lambda}$$

The S and P are conjugate foci, but there are several focal points



Homework

□ 10.32, 10.35, 10.37